

1. Phase 1: Multi-Modal Feature Matrix & Microenvironment Deconvolution (N = 326)

Analytical Methodology & Rationale

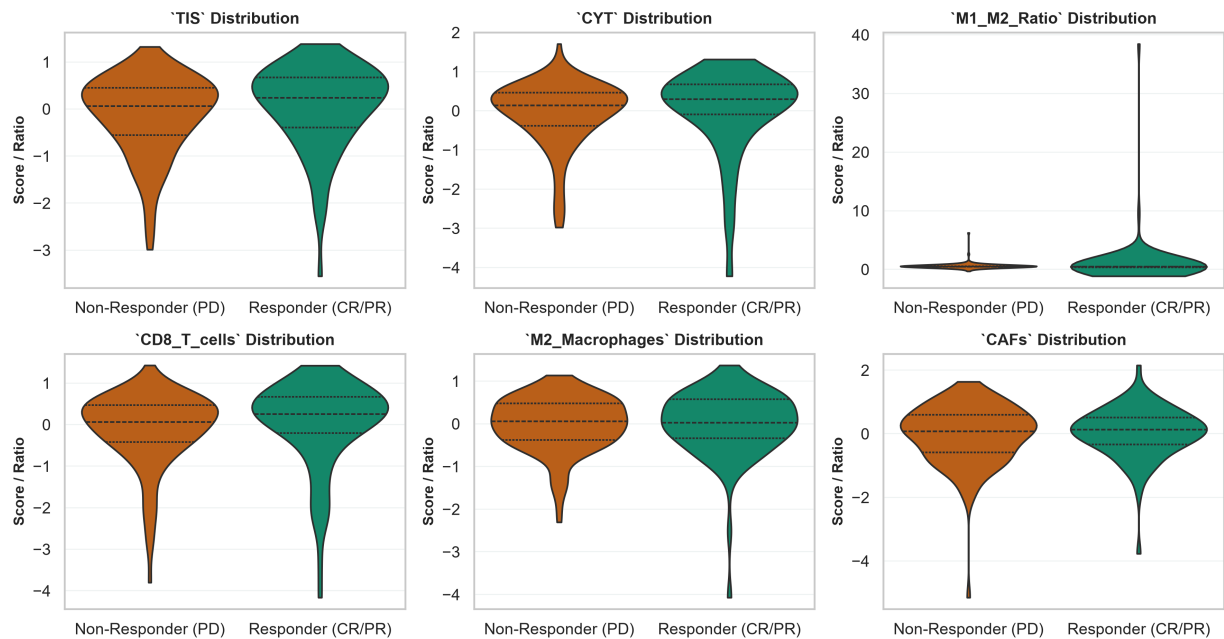
- **What is being done:** Loading preprocessed clinical, expression, and genomic data ($N = 326$) and engineering core immune signatures, Macrophage STV ratios, and cell deconvolution scores.
- **Why we are doing it:** Raw gene expression matrices containing ~19,757 genes suffer from the curse of dimensionality. Dimensionality reduction into validated signature scores and cell-type fractions provides interpretable biological features.
- **What question it answers:** What baseline immune and microenvironmental features best capture the state of tumour-infiltrating lymphocytes and immunosuppressive stroma?

Phase 1 integrates harmonised data from four clinical trials ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#), [TCGA-SKCM](#)). Rather than evaluating 19,757 genes independently, Phase 1 projects expression profiles onto curated biological axes:

- **Core Immune Signatures:** Tumour Inflammation Signature (TIS), Cytolytic Index (CYT, mean of PRF1 and GZMA), Interferon-gamma (IFN_gamma), and CD274 (PD-L1) expression.
- **Macrophage STV (M1_M2_Ratio):** Computed using a linear Signature Transcript Vector (W_g , 14,837 genes) to quantify the balance between pro-inflammatory M1 macrophages ($W_g > 0$) and pro-tumour M2 macrophages ($W_g < 0$).
- **Transcriptomic Deconvolution:** Marker-based signature scores estimating the relative abundance of CD8+ T cells, CD4+ T cells, NK cells, B cells, M1 Macrophages, M2 Macrophages, and Cancer-Associated Fibroblasts (CAFs).

Baseline Biomarker Feature Distributions

Baseline Immune & Microenvironmental Biomarker Distributions (N=195)



Key Takeaways

- **Dimensionality Reduction:** Successfully compressed ~19,757 transcriptomic features into 37 standardized, clinically interpretable biomarkers.
- **M1/M2 Polarisation:** The Macrophage STV score captures microenvironmental suppression that operates independently of total T-cell density.